

BARGUZIN, Russian FederationWMO: **306360**Lat: **53.617N**Lon: **109.633E**Elev: **489**StdP: **95.59**Time Zone: **8.00 (E08)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.7	(c) -37.8	(d) -42.5	(e) 0.1	(f) -39.6	(g) -40.5	(h) 0.1	(i) -37.7	(j) 5.9	(k) -14.9	(l) 5.2	(m) -16.9	(n) 0.1	(o) 230	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	30.1	17.8	28.1	17.4	26.3	16.7	19.6	26.0	18.6	24.9	17.7	24.2	1.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.3	21.7	16.5	12.5	20.7	15.4	11.6	20.3	58.4	26.1	55.2	24.6	52.2	24.4	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.4	(b) 4.4	(c) 3.7	(e) -42.4	(f) 33.4	(g) 1.8	(h) 2.2	(i) -43.8	(j) 34.9	(k) -44.8	(l) 36.2	(m) -45.9	(n) 37.4	(o) -47.2	(p) 39.0		
			DB													
			WB													
				(e) -42.5	(f) 20.9	(g) 1.8	(h) 1.1	(i) -43.8	(j) 21.7	(k) -44.9	(l) 22.3	(m) -45.9	(n) 22.9	(o) -47.2	(p) 23.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.6	-26.3	-21.4	-9.0	2.1	8.8	16.4	19.6	17.0	9.2	-0.2	-12.7	-23.4		
	DBStd	16.86	7.08	7.50	7.34	4.71	3.89	3.75	2.72	2.76	4.13	4.68	7.94	8.37		
	HDD10.0	5002	1126	879	588	238	71	1	0	1	64	317	681	1036		
	HDD18.3	7357	1384	1113	847	486	297	81	18	57	275	575	931	1294		
	CDD10.0	782	0	0	0	2	33	193	297	218	39	0	0	0		
	CDD18.3	96	0	0	0	0	1	22	57	16	0	0	0	0		
	CDH23.3	1227	0	0	0	0	34	372	623	187	11	0	0	0		
	CDH26.7	339	0	0	0	0	9	113	195	22	1	0	0	0		
	WSAvg	1.4	0.9	1.3	1.5	1.9	2.0	1.6	1.4	1.3	1.5	1.2	1.0	0.8		
(15) Precipitation	PrecAvg	333	13	6	5	11	14	35	74	58	39	20	34	31		
	PrecMax	512	42	19	28	29	49	109	205	132	110	54	86	74		
	PrecMin	189	1	0	0	1	0	1	18	15	6	0	11	1		
	PrecStd	77	9	5	6	7	11	27	49	33	26	12	18	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.5	0.0	9.0	20.3	27.3	32.4	32.9	28.7	25.2	14.2	6.1	1.1		
		MCWB	-7.5	-2.9	2.7	9.4	13.9	16.4	18.5	17.5	14.9	7.6	2.3	-0.8		
	2%	DB	-9.3	-3.5	6.3	15.9	23.0	29.7	30.9	27.0	21.4	10.8	3.5	-3.8		
		MCWB	-10.5	-5.2	1.2	7.0	11.8	16.8	18.7	17.1	13.1	5.7	0.9	-5.1		
	5%	DB	-12.4	-7.3	3.8	12.9	20.2	27.5	29.0	25.5	19.0	8.8	1.3	-6.2		
		MCWB	-13.3	-8.7	-0.3	5.3	10.2	16.1	18.2	16.7	12.2	4.4	-1.2	-7.2		
	10%	DB	-15.3	-10.0	1.5	10.3	17.9	25.3	27.0	23.8	16.5	6.9	-1.2	-10.3		
		MCWB	-16.1	-11.3	-1.9	3.9	8.9	15.6	17.7	16.2	10.8	3.4	-3.1	-11.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.4	-2.9	3.4	9.2	14.4	19.6	21.2	19.7	15.2	8.4	3.0	-0.5		
		MCDB	-5.8	-0.2	8.6	19.4	26.3	27.2	28.1	25.5	21.7	13.1	5.2	0.9		
	2%	WB	-10.4	-5.3	1.4	7.4	12.4	18.0	20.1	18.6	14.1	6.5	1.2	-5.0		
		MCDB	-9.3	-3.2	6.1	15.2	21.8	26.2	26.8	24.1	19.9	9.8	3.4	-3.7		
	5%	WB	-13.4	-8.8	-0.3	5.6	10.8	17.2	19.3	17.7	12.8	4.9	-1.1	-7.3		
		MCDB	-12.5	-7.6	3.5	12.1	18.7	25.3	25.7	23.4	17.8	8.3	1.2	-6.3		
	10%	WB	-16.1	-11.2	-1.8	4.1	9.3	16.3	18.5	16.8	11.5	3.3	-3.0	-11.2		
		MCDB	-15.4	-10.0	1.4	9.5	17.1	23.4	24.9	22.3	16.0	6.6	-1.1	-10.2		
(35) Mean Daily Temperature Range	MDBR		8.8	10.6	12.0	11.4	12.9	13.4	12.4	11.4	10.5	9.4	8.4	7.8		
	5% DB	MCDBR	12.7	10.3	12.6	16.5	19.0	18.2	17.0	15.2	15.2	12.6	9.3	9.7		
		MCWBR	11.7	8.6	8.1	8.7	9.3	7.2	6.8	6.7	7.4	7.6	7.3	8.8		
	5% WB	MCDBR	12.7	10.5	12.3	15.6	16.6	15.2	13.5	13.0	13.2	11.6	9.4	9.7		
		MCWBR	11.8	8.9	8.0	8.5	8.4	6.6	6.1	6.4	6.6	7.4	7.7	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.272	0.306	0.335	0.388	0.453	0.476	0.446	0.406	0.355	0.320	0.291	0.270		
	taud		2.041	2.002	2.011	2.057	2.009	2.063	2.219	2.306	2.392	2.335	2.097	2.002		
	Ebn at Noon		663	761	827	833	794	779	799	812	816	761	631	570		
	Edh at Noon		73	107	133	146	162	156	131	113	92	77	69	62		
(44) All-Sky Solar Radiation	RadAvg		0.97	1.96	3.37	4.71	5.67	6.09	5.55	4.64	3.34	1.92	0.94	0.65		
	RadStd		0.08	0.12	0.14	0.30	0.35	0.37	0.48	0.22	0.27	0.19	0.10	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	-1.82	-2.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page